

Multiple Choice (10 marks)

1. Statement 1| As of 2020, some models attain greater than 98% accuracy on CIFAR-10.
Statement 2| The original ResNets were not optimized with the Adam optimizer.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Predicting the amount of rainfall in a region based on various cues is a _____ problem.
 - (a) Supervised learning
 - (b) Unsupervised learning
 - (c) Clustering
 - (d) None of the above
3. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
 - (a) Increase the amount of training data.
 - (b) Improve the optimisation algorithm being used for error minimisation.
 - (c) Decrease the model complexity.
 - (d) Reduce the noise in the training data.
4. What would you do in PCA to get the same projection as SVD?
 - (a) Transform data to zero mean
 - (b) Transform data to zero median
 - (c) Not possible
 - (d) None of these
5. High entropy means that the partitions in classification are
 - (a) pure
 - (b) not pure
 - (c) useful
 - (d) useless

True / False (10 marks)

Answer True or False

6. True or False: The computational complexity of gradient descent is polynomial in the number of dimensions, D .
7. True or False: Out-of-distribution detection is commonly referred to as background detection in the field of machine learning.
8. True or False: A Bayesian network with the structure $H \rightarrow U \leftarrow P \leftarrow W$ requires 4 independent parameters to fully define its conditional probabilities.
9. True or False: K -fold cross-validation has a computational complexity that is linear in K , as it involves training the model K times.
10. True or False: The cost of one gradient descent update given the gradient is $O(N)$, where N is the number of training examples.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to remove odd characters in a string.
12. Write a function to check whether the given number is armstrong or not.

End of Paper